PRACTICAL TRAINING REPORT

CARS-SF: A Context-Aware Autonomous Rural Service Slicing Framework for Private 5G Networks

Submitted in partial fulfillment of the requirements
for the Summer Internship

Submitted By

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Academic Year 2026–2027

Abstract

Rural and remote communities depend on communication networks for healthcare, education, agriculture, emergency response, and other essential services. These services have different network requirements and may need different priorities as the operating situation changes. Conventional network slicing can provide different resource allocations, but fixed policies cannot easily respond when these priorities change.

This work presents CARS-SF, a Context-Aware Autonomous Rural Service Slicing Framework for private 5G networks. The proposed framework aims to identify incoming traffic using SliceNet, combine the identified service with the current operating context and network condition, and use this information to make adaptive resource-allocation decisions through reinforcement learning. Network monitoring provides feedback so that future decisions can respond to changing conditions.

The internship focused on establishing the foundation for this framework. A configurable Mininet-based rural network environment was developed along with service-aware operational scenarios and a hybrid traffic-generation pipeline. The Rural Service-aware Network Slice Dataset (RSNSD) was created by combining network-flow characteristics with service and scenario information, and an initial SliceNet classification framework was established using this dataset. These components provide the simulation, data, and traffic-classification foundation for the subsequent development of adaptive resource management and autonomous network control.

Keywords: Private 5G, Rural Networks, Network Slicing, Service-aware Networking, Context-aware Networking, Reinforcement Learning, SliceNet, RSNSD, Autonomous Networks.

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Chapter 1

Introduction

Today, many applications depend on a stable Internet connection, but all applications do not have the same requirements. This becomes more important in rural areas, where a common network may support online education, healthcare, agriculture, general Internet access and emergency communication. During normal conditions, these services can share the available resources, but during an emergency, critical services such as healthcare and emergency communication should receive higher priority.

Traditional networks mainly use fixed QoS rules and predefined resource allocation, which does not adapt well when traffic patterns or operating conditions changes. 5G network slicing and Software Defined Networking (SDN) provide mechanisms to separate and control resources for different services, but their decisions are still largely based on predefined policies. Machine Learning can identify the type of incoming traffic, but traffic classification alone cannot determine how important that traffic is under the current situation. For example, healthcare traffic may require higher priority during an emergency, while educational traffic may become more important during an examination or learning session. Hence, the network needs to understand both the traffic and the context in which it is operating.

From the study of existing approaches, we identified that traffic classification, network slicing and intelligent resource allocation are often handled as separate problems, with limited integration between them. This led to the research gap for a framework that can combine these capabilities into a single closed-loop system. Therefore, this work proposes an AI-driven, context-aware 5G network framework in which incoming traffic is classified using SliceNet, contextual information is used to determine service priority, and a Reinforcement Learning (RL) agent dynamically adapts network resources according to the changing network conditions. The proposed approach aims to move from fixed resource allocation towards an adaptive network that can identify which services are important in the current scenario and provide them with better connectivity when required.

Chapter 2

Problem Motivation

2.1 Rural Mission-Critical Communication Scenario

Rural areas depend on communication networks for many important services, including healthcare, education, agriculture, emergency services, and general Internet access. These services share the same network, but they do not have the same needs. For example, healthcare and emergency communication may need low delay and reliable delivery, education may need more bandwidth during school hours, while general Internet traffic can usually work with lower priority.

Table 2.1: Representative rural services and their changing network requirements

Service	**Typical Requirement**	**Context Affecting Priority**	**Expected Network Behaviour**
Healthcare	Low delay, reliability	Normal / Emergency / Disaster	Higher priority during critical events
Education	Sustained bandwidth	School Hours	More resources during peak activity
Agriculture	Periodic sensing and IoT traffic	Harvest Season	Support increased monitoring demand
Emergency	Low delay, high reliability	Emergency / Disaster	Highest priority during critical conditions
General Internet	Best-effort access	All scenarios	Use remaining available resources

The needs of these services can also change with the situation. During normal operation, network resources can be shared among different services. During an emergency or disaster, however, healthcare and emergency communication may become more important, while general Internet traffic may have to use fewer resources. Similarly, education or agriculture may need more resources during periods of high activity. The main chal-

lenge is therefore that service priority cannot always be fixed in advance; it must change according to the service, the current situation, and the condition of the network.

This creates the need for a network that can understand what type of traffic it is carrying, recognize the current situation, and adjust its resource allocation accordingly. Private 5G provides a suitable environment for this because its network resources and service policies can be configured according to application needs.

2.2 Operational Challenge

Rural networks may have limited resources while the type and amount of traffic can change with time and operating conditions. Network slicing allows different services to share the same infrastructure with different resource allocations [1], but fixed slice policies cannot automatically respond when service priorities change. For example, during an emergency, healthcare and emergency communication may require more resources, while general Internet traffic may need to operate with lower priority.

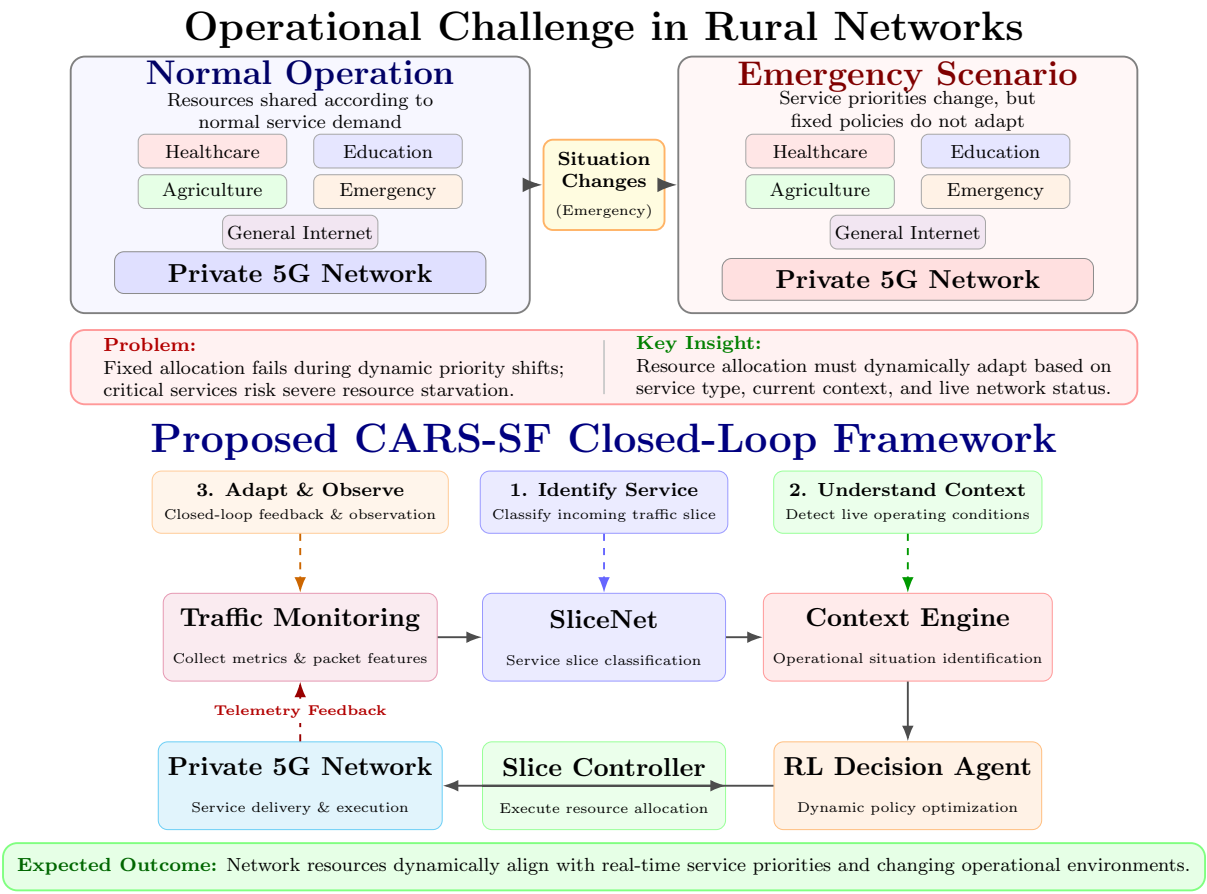


Figure 2.1: Operational challenge in rural networks and the proposed CARS-SF closed-loop framework for service-aware and context-aware resource management.

Standard 5G service categories such as eMBB, URLLC, and mMTC provide broad descriptions of communication needs, but do not directly identify the application gener-

ating the traffic or its importance in a specific situation. The network therefore needs to understand both the service being carried and the current operating condition before deciding how resources should be distributed. Research on intelligent slicing and reinforcement learning has already explored dynamic resource allocation under changing network conditions.

2.3 Research Problem and Identified Research Gap

This creates three basic requirements for CARS-SF: the network must identify the service generating the traffic, understand the current situation, and adjust its resources accordingly. It must also monitor the result so that future decisions can respond to changes in the network [8]. Therefore, the main problem is how to make network resource allocation change according to the service, the current situation, and the network condition.

Based on this, the research problem addressed in this work is:

> *How can a private 5G network serving different rural services identify the service generating the traffic, understand the current situation, and adjust network resources according to the changing needs of those services?*

Existing research has already addressed important parts of this problem. Network slicing allows different services to share the same infrastructure with different resource and QoS requirements [1]. Machine-learning-based traffic classification can identify the type of service from network traffic. Reinforcement learning has also been studied for changing resource allocation and network-slice management [5, 6, 7]. These works show that the individual components required for an adaptive network are possible. However, they do not by themselves provide the complete service–context–decision–feedback loop required for the rural scenario considered in this work.

The gap identified from this study is therefore mainly an **integration gap**. The network needs to know *what service is being carried*, *what is happening in the current situation*, and *what the network condition is* before deciding how resources should be allocated. After the action is applied, the network must also observe its effect and use that information for the next decision. Recent work has demonstrated real-time service-aware slice adaptation using traffic classification and network information [**?**], but the CARS-SF problem further considers changing service priority based on the operational situation of the rural network.

Table 2.2: Literature-derived capabilities and resulting CARS-SF design requirements

Capability	**Existing Contribution**	**Requirement for CARS-SF**
Network Slicing	Logical isolation and different QoS over shared infrastructure [1]	Dynamic service-level resource allocation
Service-aware Slicing	Considers application requirements during slice management	Service information in resource decisions
Traffic Classification	Identifies service or application characteristics from network flows [3, 4]	Service perception through SliceNet
RL-based Management	Adapts resource allocation under changing network conditions [5, 7]	Context-aware adaptive decisions
Autonomous Networking	Uses monitoring and feedback for network adaptation [8]	Continuous closed-loop operation

CARS-SF is therefore designed to connect these capabilities within one framework. SliceNet provides information about the service represented by incoming traffic, the context layer represents the current operating situation, and the reinforcement learning component provides the adaptive resource-management decision. Network monitoring provides the feedback required to evaluate each decision. The framework consequently moves from simply assigning traffic to predefined slices towards resource allocation that can respond to the service, situation, and condition of the network.

2.4 Research Objective

The objective of CARS-SF is to develop a framework that can manage network resources according to the service being carried and the current operating situation in a rural private 5G network. The framework aims to identify incoming traffic, understand its importance in the current situation, adjust network resources accordingly, and monitor the result. This creates a continuous cycle in which network decisions can change as the traffic and operating conditions change.

The internship focuses on establishing the foundation for this framework. This includes developing a configurable Mininet-based rural network environment, modelling different services and operating scenarios, creating the Rural Service-aware Network Slice Dataset (RSNSD), and developing the initial SliceNet traffic-classification framework.

Chapter 3

Background and Literature Study

3.1 Private 5G for Rural and Mission-Critical Applications

Private 5G networks provide greater control over network resources, Quality of Service (QoS), security, and service management than conventional public networks. This makes them suitable for environments where different services share the same network but have different communication requirements.

This is particularly relevant in rural areas, where a common network may support healthcare, education, agriculture, public services, and emergency communication. Previous work has also studied network slicing as a way to support multiple rural services over shared 5G infrastructure, while 5G-based communication has been explored for applications such as agriculture.

For CARS-SF, private 5G therefore provides the network environment in which service-specific policies and resource allocation can be controlled and adapted. It forms the underlying platform for the service-aware and context-aware resource management proposed in this work.

3.2 Network Slicing and Service Differentiation

Network slicing allows different services to share the same 5G infrastructure while receiving different resource and Quality of Service (QoS) requirements [1]. Standard categories such as eMBB, URLLC, and mMTC provide broad service classifications, but rural networks may need to distinguish between specific applications such as healthcare, education, agriculture, and emergency communication. Service-aware slicing therefore considers the requirements of individual services when managing network resources

For CARS-SF, this means that the network must first know which service is generating the traffic before deciding how its resources should be managed. This creates the need

for a traffic-classification layer that can identify services from observed network traffic.

Table 3.1: Selected literature on intelligent traffic classification and adaptive network resource management

Work	**Area**	**Main Contribution**	**Relevance to CARS-SF**
Xie *et al.* [3]	Traffic Classification	Uses machine learning to classify traffic associated with 5G network slices.	Supports service-aware traffic perception.
Mohammedali *et al.* [4]	Traffic Classification	Applies deep learning for traffic classification in end-to-end 5G/B5G slice management.	Connects traffic classification with slice management.
Li *et al.* [5]	Reinforcement Learning	Uses deep reinforcement learning for dynamic resource management in network slicing.	Provides the basis for adaptive resource decisions.
Yan *et al.* [6]	RAN Resource Management	Applies intelligent resource scheduling to 5G RAN slicing.	Supports dynamic resource allocation.
Sciancalepore *et al.* [7]	Slice Management	Uses reinforcement learning for network-slice broker decisions.	Supports adaptive slice-level control.
Moysen and Giupponi [8]	Autonomous Networking	Studies machine learning for self-organized network management.	Supports the monitoring and adaptation concept.

3.3 Intelligent Traffic Perception and Autonomous Resource Management

Machine-learning-based traffic classification can identify the service represented by network traffic using observable flow characteristics [3, 4]. For CARS-SF, this forms the perception stage, where SliceNet converts incoming network observations into service information that can be used by the network-management system.

Reinforcement learning provides a way to make resource-allocation decisions based on the current network condition instead of relying only on fixed rules. Previous studies have explored RL for dynamic resource allocation and network-slice management under changing traffic conditions [5, 7]. In CARS-SF, the RL agent is intended to use service information, operating context, and network conditions to decide how available resources

should be adjusted.

These two capabilities can be combined with continuous network monitoring to form a closed loop. The network observes traffic, identifies the service, makes a resource-allocation decision, applies the change, and observes the resulting network behaviour. This follows the broader idea of autonomous network management, where monitoring and adaptation reduce dependence on fixed policies [8]. This closed-loop approach forms the basis of the CARS-SF framework and its future extension towards self-healing network operation.

Chapter 4

Internship Engineering Work

The internship focused on establishing the experimental and data foundation required for the CARS-SF framework. The work resulted in a reproducible rural network environment, representative service and operational scenarios, a service-aware network dataset, and the initial traffic-classification pipeline.

4.1 Experimental Environment and Scenario Modeling

A configurable Mininet-based network environment was developed to emulate the communication characteristics of a rural deployment without requiring dedicated physical 5G hardware. The environment incorporates service-specific hosts, Open vSwitch-based forwarding, configurable network links, traffic generation, and network monitoring, providing a controlled platform for generating and observing heterogeneous service traffic.

The environment was configured to represent different operational conditions in which service demand and relative priorities can change. The implemented scenarios include normal operation, increased education-related activity during school hours, elevated agricultural and IoT activity during the harvest season, emergency operation with greater priority for critical services, and disaster recovery under constrained resource availability. These scenarios provide the contextual variation required for evaluating service-aware network management.

4.2 RSNSD: Rural Service-aware Network Slice Dataset

To support service-aware traffic analysis, a hybrid traffic-generation and data-collection pipeline was established by combining simulation-derived network traffic with statistically generated synthetic traffic. The resulting Rural Service-aware Network Slice Dataset (RSNSD) captures network-flow characteristics together with service and operational

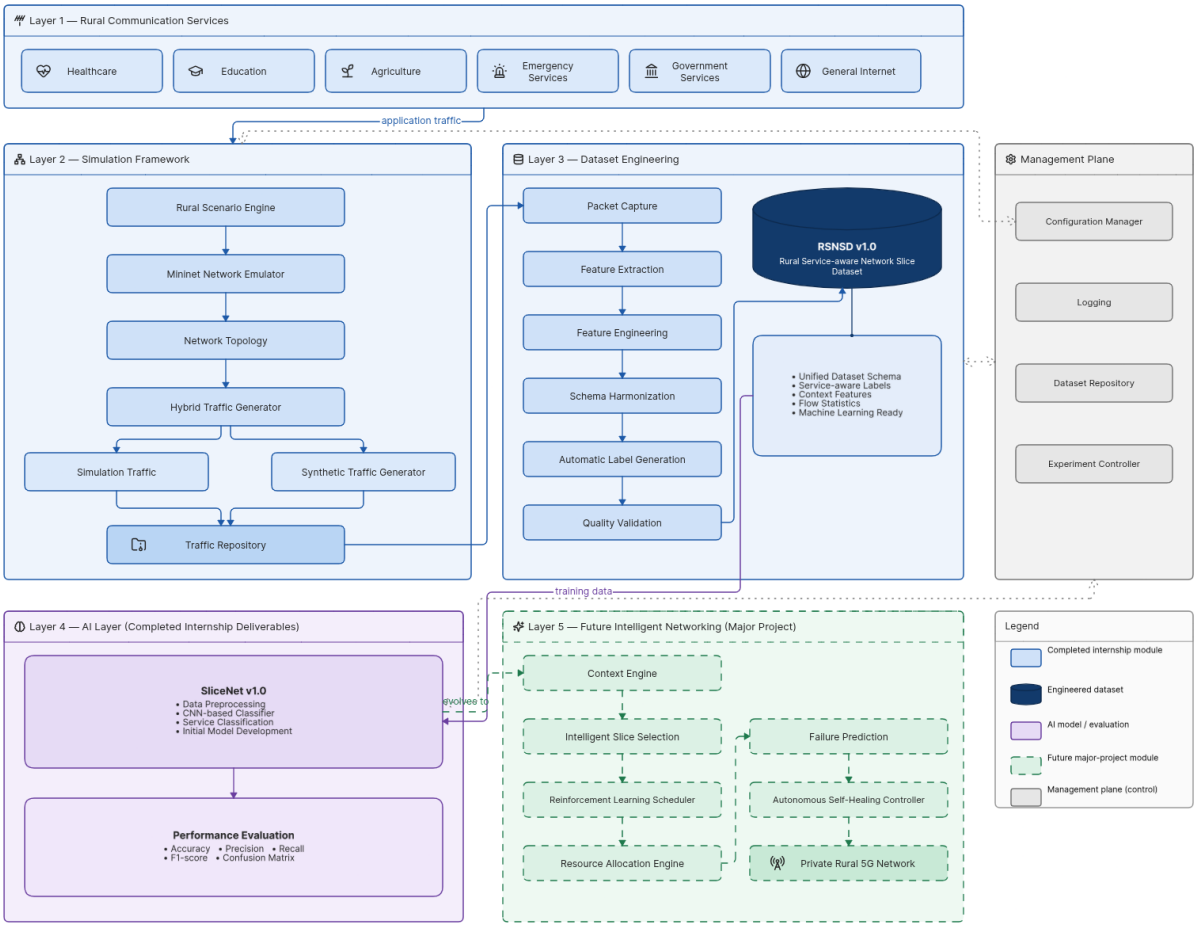


Figure 4.1: Dataset engineering and data flow architecture

context, providing a unified data foundation for the machine-learning components of CARS-SF.

The dataset incorporates representative Layer 2–4 features, flow and temporal statistics, service labels, operational scenarios, and associated QoS priorities. This representation allows network observations to be related not only to their measurable traffic characteristics, but also to the service and context they represent.

4.3 SliceNet Engineering Foundation

SliceNet was formulated as the service-perception component of CARS-SF, using RSNSD to frame traffic classification as a supervised multi-class learning problem. The internship established the classification pipeline, including feature representation, preprocessing, model formulation, and the interface required for integration with subsequent network-management components.

The resulting environment, dataset, and SliceNet foundation establish the perception and experimentation layers of CARS-SF. The integration of this perception layer with context-aware resource orchestration and reinforcement learning-based network control

forms the subsequent development direction of the major project.

Table 4.1: Internship deliverables and their role in the CARS-SF framework

Deliverable	**Engineering Work Established**	**Role in CARS-SF**
Mininet Rural Environment	Configurable topology, service-specific hosts, OVS forwarding, traffic generation, and monitoring	Experimental foundation
Operational Scenarios	Normal operation, school hours, harvest season, emergency, and disaster recovery	Context representation
Traffic Generation Pipeline	Simulation-derived and synthetic service traffic	Representative network observations
RSNSD	L2–L4, flow, temporal, service, context, and QoS information	Machine-learning data foundation
SliceNet Foundation	Feature representation, preprocessing, and supervised service-classification pipeline	Service-perception layer

Chapter 5

Conclusion and Future Direction

The internship established the main engineering components required for the CARS-SF framework. A configurable Mininet-based rural network environment was developed to represent different services and operating conditions. It supports service-specific traffic generation, configurable network settings, and network monitoring, providing the environment needed for further experiments.

The Rural Service-aware Network Slice Dataset (RSNSD) was then developed using simulation-based and synthetic traffic. The dataset links network flow features with service classes, operating scenarios, and QoS priorities, providing the data required for service-aware traffic classification.

The initial SliceNet framework was also established using RSNSD to classify incoming traffic into service categories. Thus, the internship delivered three main outcomes: the rural network simulation environment, the RSNSD dataset, and the initial SliceNet classification pipeline. Together, these provide the simulation, data, and service-classification base required for the next stage of CARS-SF.

The remaining major-project work will build on these components by integrating them with the private 5G environment, context-aware resource management, and reinforcement learning-based control. The final system will be evaluated under changing service demands and operating conditions, with dynamic resource adaptation and self-healing forming the longer-term direction of the framework.

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